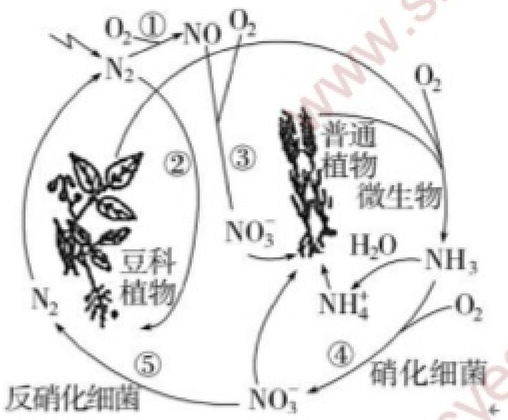
第四单元 化学实验与应用（Unit 4 Chemistry Experiments and Applications）

第四章 工业化工流程分析（如合成氨）（Chapter 4 Analysis of Industrial Chemical Processes (e.g., Ammonia Synthesis)）

1.下列关于自然界中氮循环 (如图) 的说法错误的是 (Which of the following statements about the nitrogen cycle in nature (as shown in the figure) is incorrect?) ( )



A.①和②均属于氮的固定 (① and ② both belong to nitrogen fixation)

B. 氢、氧元素也参与了自然界中的氮循环 (Hydrogen and oxygen elements also participate in the nitrogen cycle in nature)

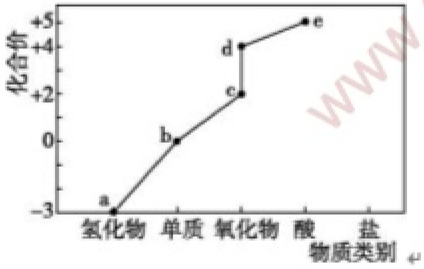
C.④中每生成 1 mol NO_3^-，消耗 2 mol O_2(2 mol of O_2 is consumed for every 1 mol of NO_3^- generated in ④)

D.③和⑤过程中氮元素分别被还原和被氧化 (Nitrogen is reduced and oxidized in processes ③ and ⑤ respectively)

答案 (Answer):D

解析 (Analysis)：氮的固定指将 N_2转化为氮的化合物，反应①将 N_2转化为 NO，属于氮的固定，反应②属于自然固氮，A 正确；由图示知，氮循环过程中涉及氢、氧元素，B 正确；反应④是 O_2将 NH_3氧化为 NO_3^-，N 元素化合价由 - 3 价升高到 + 5 价，则生成 1 mol NO_3^-失去 8 mol 电子，失去的电子被 O_2得到，根据转移电子关系 O_2~$4e^-$，可知得 8 mol 电子需要 2 mol O_2，C 正确；反应③中 NO 转化为 NO_3^-，N 的化合价升高，氮元素被氧化，反应⑤中 NO_3^-转化为 N_2，N 的化合价降低，氮元素被还原，D 错误。(Nitrogen fixation refers to the conversion of N_2 into nitrogen compounds. Reaction ① converts N_2 into NO, which belongs to nitrogen fixation; reaction ② belongs to natural nitrogen fixation, so A is correct. As shown in the figure, hydrogen and oxygen elements are involved in the nitrogen cycle, so B is correct. Reaction ④ is the oxidation of NH_3 to NO_3^- by O_2. The valence of N increases from -3 to +5, so 8 mol of electrons are lost when 1 mol of NO_3^- is generated. These electrons are gained by O_2. According to the electron transfer relationship O_2 ~ $4e^-$, 2 mol of O_2 is needed to gain 8 mol of electrons, so C is correct. In reaction ③, NO is converted to NO_3^-, the valence of N increases, and nitrogen is oxidized; in reaction ⑤, NO_3^- is converted to N_2, the valence of N decreases, and nitrogen is reduced, so D is incorrect.)

2.部分含氮物质的分类与相应化合价关系如图所示。下列推断不合理的是 (The classification and corresponding valence relationship of some nitrogen-containing substances are shown in the figure. Which of the following inferences is unreasonable?) ( )

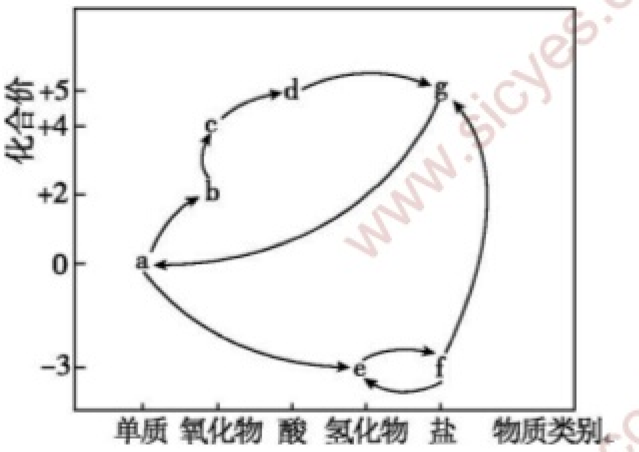


A.a 可经催化氧化生成 c (a can be catalytically oxidized to form c)
B.b 既可被氧化，也可被还原 (b can be both oxidized and reduced)
C.d 与水反应只生成 e (d reacts with water to form only e)
D.“雷雨发庄稼” 包括 b→c→d→e 的转化 (Thunderstorms nourish crops includes the transformation of b→c→d→e)

答案 (Answer):C

解析 (Analysis)：部分含氮物质的分类与相应化合价关系如图所示，分析价态和物质分类得到，a 为 NH_3，b 为 N_2，c 为 NO，d 为 NO_2，e 为 HNO_3。氨气和氧气在催化剂、加热条件下反应生成 NO 和水，NH_3可经催化氧化生成 NO，A 正确；N_2中氮元素化合价为 0 价，可以升高，也可以降低，既可被氧化，也可被还原，B 正确；NO_2与水反应生成 HNO_3和 NO，C 错误；“雷雨发庄稼”：氮气先与氧气在闪电条件下反应生成 NO，继续被氧化为二氧化氮，再与水反应生成硝酸，包括 N_2→NO→NO_2→HNO_3的转化，D 正确。(The classification and corresponding valence relationship of some nitrogen-containing substances are shown in the figure. By analyzing the valence and substance classification, a is NH_3, b is N_2, c is NO, d is NO_2, and e is HNO_3. Ammonia reacts with oxygen to form NO and water under the conditions of catalyst and heating, so NH_3 can be catalytically oxidized to form NO, so A is correct. The valence of nitrogen in N_2 is 0, which can be increased or decreased, so it can be both oxidized and reduced, so B is correct. NO_2 reacts with water to form HNO_3 and NO, so C is incorrect. Thunderstorms nourish crops: Nitrogen first reacts with oxygen to form NO under lightning conditions, is further oxidized to nitrogen dioxide, and then reacts with water to form nitric acid, which includes the transformation of N_2→NO→NO_2→HNO_3, so D is correct.)

3.自然界中氮的循环意义重大。其中，部分含氮物质的转化关系呈现在氮及其化合物的 “价 — 类” 二维图中。下列说法不正确的是 (The nitrogen cycle in nature is of great significance. Among them, the transformation relationship of some nitrogen-containing substances is shown in the valence-category two-dimensional diagram of nitrogen and its compounds. Which of the following statements is incorrect?) ( )



A.a→b 的转化是工业制硝酸的基础 (The transformation of a→b is the basis for industrial production of nitric acid)

B. 既属于 g 又属于 f 的盐可用作氮肥 (The salt belonging to both g and f can be used as nitrogen fertilizer)

C. 汽车中的三元催化器可促进尾气中的 b、c 向 a 转化 (The three-way catalyst in automobiles can promote the transformation of b and c in exhaust gas to a)

D. 实验室中，常用 f 与强碱反应产生 e 来检验 f 中阳离子的存在 (In the laboratory, f is usually reacted with strong base to produce e to test the existence of cations in f)

答案 (Answer):A

解析 (Analysis)：由 N 元素的化合价及物质所属类别，可确定 a 为 N_2、b 为 NO、c 为 NO_2、d 为 HNO_3、e 为 NH_3、f 为铵盐、g 为硝酸盐。工业制硝酸的基础是氨的催化氧化，即 NH_3→NO，而 a 为 N_2，则 N_2→NO 的转化不是工业制硝酸的基础，A 不正确；既属于 g (硝酸盐) 又属于 f (铵盐) 的盐为 NH_4NO_3，它含氮量较高，可用作氮肥，B 正确；NO、NO_2是大气污染物，利用汽车中的三元催化器，可促进尾气中 NO、NO_2与 CO 作用，生成 N_2和 CO_2，C 正确；实验室中，检验 NH_4^+时，常用铵盐与强碱在加热条件下反应，通过检验反应产生的 NH_3以验证 NH_4^+的存在，D 正确。(According to the valence of N element and the category of substances, a is N_2, b is NO, c is NO_2, d is HNO_3, e is NH_3, f is ammonium salt, and g is nitrate. The basis for industrial production of nitric acid is the catalytic oxidation of ammonia, namely NH_3→NO, while a is N_2, so the transformation of N_2→NO is not the basis for industrial production of nitric acid, so A is incorrect. The salt belonging to both g (nitrate) and f (ammonium salt) is NH_4NO_3, which has a high nitrogen content and can be used as nitrogen fertilizer, so B is correct. NO and NO_2 are air pollutants. The three-way catalyst in automobiles can promote the reaction of NO and NO_2 in exhaust gas with CO to form N_2 and CO_2, so C is correct. In the laboratory, when testing NH_4^+, ammonium salt is usually reacted with strong base under heating conditions, and the existence of NH_4^+ is verified by testing the NH_3 produced by the reaction, so D is correct.)

4.氮及其化合物的转化具有重要应用，下列说法不正确的是 (The transformation of nitrogen and its compounds has important applications. Which of the following statements is incorrect?) ( )

A. 工业上模拟自然界 “雷雨发庄稼” 的过程生产 HNO_3(Industry simulates the process of thunderstorms nourish crops in nature to produce HNO_3)

B. 自然固氮、人工固氮都是将 N_2转化为含氮化合物 (Both natural nitrogen fixation and artificial nitrogen fixation convert N_2 into nitrogen-containing compounds)

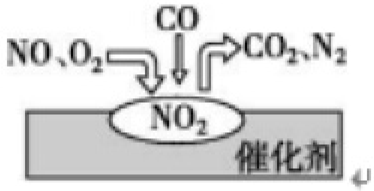
C. 氨气是重要的工业原料，可用于制备化肥和纯碱等大宗化学品 (Ammonia is an important industrial raw material, which can be used to prepare bulk chemicals such as chemical fertilizers and soda ash)

D. 多种形态的氮及其化合物间的转化形成了自然界的 “氮循环”(The transformation between various forms of nitrogen and its compounds forms the nitrogen cycle in nature)

答案 (Answer):A

解析 (Analysis)：自然界 “雷雨发庄稼” 的第一步是将空气中的氮气转化为 NO，工业生产硝酸是氨气经催化氧化得到 NO，NO 被氧化为 NO_2，NO_2和水反应最后得到硝酸，A 项错误；氮的固定是指将 N_2转化为含氮化合物的过程，其中氮的固定包括自然固氮、人工固氮，B 项正确；氨气作为重要的工业原料可以制备化肥和纯碱等大宗化学品，C 项正确；氮元素在自然界中既有游离态又有化合态，多种形态的氮及其化合物间的转化形成了自然界的 “氮循环”，D 项正确。(The first step of thunderstorms nourish crops in nature is to convert nitrogen in the air into NO. Industrial production of nitric acid is to obtain NO by catalytic oxidation of ammonia, oxidize NO to NO_2, and finally react NO_2 with water to obtain nitric acid, so A is incorrect. Nitrogen fixation refers to the process of converting N_2 into nitrogen-containing compounds, including natural nitrogen fixation and artificial nitrogen fixation, so B is correct. As an important industrial raw material, ammonia can be used to prepare bulk chemicals such as chemical fertilizers and soda ash, so C is correct. Nitrogen exists in both free state and combined state in nature, and the transformation between various forms of nitrogen and its compounds forms the nitrogen cycle in nature, so D is correct.)

5.汽车尾气处理装置中，气体在催化剂表面吸附与解吸的过程如图所示，下列说法正确的是 (The process of gas adsorption and desorption on the catalyst surface in the automobile exhaust treatment device is shown in the figure. Which of the following statements is correct?) ( )



A. 汽车尾气的主要污染成分包括 CO、NO 和 N_2(The main pollutants of automobile exhaust include CO, NO and N_2)

B. 反应中 NO 为氧化剂，N_2为还原产物 (NO is the oxidizing agent and N_2 is the reduction product in the reaction)

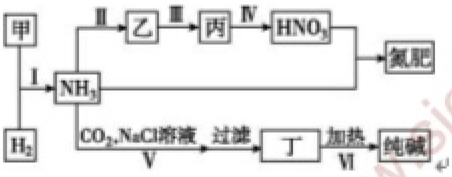
[C.NO](C.NO) 和 O_2必须在催化剂表面才能反应 (NO and O_2 must react on the catalyst surface)

D. 催化转化总反应为 $2NO+O_2+4CO \overset{催化剂}{=} 4CO_2+N_2$(The total catalytic conversion reaction is $2NO+O_2+4CO \overset{catalyst}{=} 4CO_2+N_2$)

答案 (Answer):D

解析 (Analysis)：汽车尾气的主要污染物有 CO、NO 等，N_2不是污染物，A 错误；由图可知，NO、O_2、CO 在催化剂表面反应生成 CO_2和 N_2，NO、O_2是氧化剂，N_2是还原产物，B 错误；NO 与空气中 O_2相遇即可发生反应生成 NO_2，不需要催化剂，C 错误；NO、O_2、CO 在催化剂表面反应生成 CO_2和 N_2，结合守恒规律写出化学方程式：$2NO+O_2+4CO \overset{催化剂}{=} 4CO_2+N_2$，D 正确。(The main pollutants of automobile exhaust include CO, NO, etc., and N_2 is not a pollutant, so A is incorrect. As shown in the figure, NO, O_2 and CO react on the catalyst surface to form CO_2 and N_2. NO and O_2 are oxidizing agents, and N_2 is the reduction product, so B is incorrect. NO reacts with O_2 in the air to form NO_2 without a catalyst, so C is incorrect. NO, O_2 and CO react on the catalyst surface to form CO_2 and N_2. The chemical equation is written according to the conservation law: $2NO+O_2+4CO \overset{catalyst}{=} 4CO_2+N_2$, so D is correct.)

6.合成氨及其相关工业中，部分物质间的转化关系如图所示。下列说法不正确的是 (The transformation relationship of some substances in synthetic ammonia and its related industries is shown in the figure. Which of the following statements is incorrect?) ( )



A. 甲、乙、丙三种物质中都含有氮元素 (Substances A, B and C all contain nitrogen)

B. 反应 Ⅱ、Ⅲ 和 Ⅳ 的氧化剂一定相同 (The oxidizing agents of reactions Ⅱ, Ⅲ and Ⅳ must be the same)

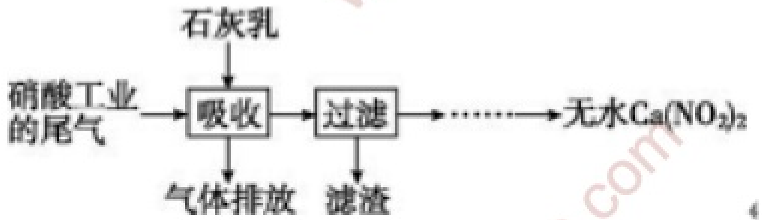
C.Ⅵ 的产物可在上述流程中被再次利用 (The product of Ⅵ can be reused in the above process)

D.Ⅴ 中发生反应：$NH_3+CO_2+H_2O+NaCl=NaHCO_3\downarrow+NH_4Cl$ (The reaction in Ⅴ is: $NH_3+CO_2+H_2O+NaCl=NaHCO_3\downarrow+NH_4Cl$)

答案 (Answer):B

解析 (Analysis)：甲是 N_2，与 H_2化合生成 NH_3；反应 Ⅴ 是侯氏制碱法，丁是 $NaHCO_3$，受热分解生成 Na_2CO_3；NH_3发生催化氧化生成 NO，NO 与 O_2反应生成 NO_2，NO_2溶于水生成 HNO_3，HNO_3 和 NH_3反应生成 NH_4NO_3，即乙是 NO，丙是 NO_2。甲、乙、丙三种物质中都含有氮元素，A 正确；反应 Ⅳ 为 $3NO_2+H_2O=2HNO_3+NO$，氧化剂和还原剂均是 NO_2，B 错误；$NaHCO_3$分解生成的 CO_2可以循环利用，C 正确；Ⅴ 中发生反应：$NH_3+CO_2+H_2O+NaCl=NaHCO_3\downarrow+NH_4Cl$，D 正确。(A is N_2, which combines with H_2 to form NH_3; reaction Ⅴ is Hou's process for soda production, D is $NaHCO_3$, which decomposes into Na_2CO_3 when heated; NH_3 is catalytically oxidized to form NO, NO reacts with O_2 to form NO_2, NO_2 dissolves in water to form HNO_3, and HNO_3 reacts with NH_3 to form NH_4NO_3, that is, B is NO and C is NO_2. Substances A, B and C all contain nitrogen, so A is correct. Reaction Ⅳ is $3NO_2+H_2O=2HNO_3+NO$, and both the oxidizing agent and reducing agent are NO_2, so B is incorrect. CO_2 produced by the decomposition of $NaHCO_3$ can be recycled, so C is correct. The reaction in Ⅴ is: $NH_3+CO_2+H_2O+NaCl=NaHCO_3\downarrow+NH_4Cl$, so D is correct.)

7.利用硝酸工业的尾气 (含 NO、NO_2) 获得 Ca $(NO_2)_2$的部分工艺流程如下。下列说法不正确的是 (The partial process flow of obtaining Ca $(NO_2)_2$ from tail gas (containing NO, NO_2) in nitric acid industry is as follows. Which of the following statements is incorrect?) ( )



A. 吸收硝酸工业尾气的石灰乳不能用澄清石灰水替换 (Milk of lime for absorbing tail gas from nitric acid industry cannot be replaced by clear lime water)

B. 为使尾气中 NO_x被充分吸收，尾气与石灰乳采用气液逆流接触吸收 (To fully absorb NO_x in tail gas, tail gas and milk of lime are absorbed by gas-liquid countercurrent contact)

C. 若尾气中，n (NO_2) : n (NO) < 1 : 1，Ca $(NO_2)_2$产品中 Ca $(NO_3)_2$含量升高 (If n (NO_2) : n (NO) < 1 : 1 in tail gas, the content of Ca $(NO_3)_2$ in Ca $(NO_2)_2$ product increases)

D.$Ca(NO_2)_2$在酸性溶液中分解的离子方程式为 $3NO_2^-+2H^+=NO_3^-+2NO\uparrow+H_2O$ (The ionic equation for the decomposition of Ca $(NO_2)_2$ in acidic solution is $3NO_2^-+2H^+=NO_3^-+2NO\uparrow+H_2O$)

答案 (Answer):C

解析 (Analysis)：吸收硝酸工业尾气的石灰乳不能用澄清石灰水替换，因为石灰水中溶质太少，不能很好地吸收尾气，故 A 正确；为使尾气中 NO_x被充分吸收，尾气与石灰乳采用气液逆流，使得接触面积增大，能充分吸收，故 B 正确；反应过程中有①$NO+NO_2+Ca(OH)_2=Ca(NO_2)_2+H_2O$、②$4NO_2+2Ca(OH)_2=Ca(NO_3)_2+Ca(NO_2)_2+2H_2O$，若尾气中，n (NO_2) : n (NO) < 1 : 1，说明一氧化氮比二氧化氮多，则尾气中 NO 含量较多，以反应①为主，产品中 Ca $(NO_2)_2$含量更多，但当 n (NO_2) : n (NO) > 1 : 1 时，NO_2含量较多，产品中 Ca $(NO_3)_2$含量升高，故 C 错误；Ca $(NO_2)_2$在酸性溶液中分解，亚硝酸根离子发生氧化还原反应即歧化反应，其离子方程式为 $3NO_2^-+2H^+=NO_3^-+2NO\uparrow+H_2O$，故 D 正确。(Milk of lime for absorbing tail gas from nitric acid industry cannot be replaced by clear lime water, because the solute in lime water is too little to absorb tail gas well, so A is correct. To fully absorb NO_x in tail gas, tail gas and milk of lime adopt gas-liquid countercurrent to increase the contact area and achieve full absorption, so B is correct. There are reactions ① $NO+NO_2+Ca(OH)_2=Ca(NO_2)_2+H_2O$ and ② $4NO_2+2Ca(OH)_2=Ca(NO_3)_2+Ca(NO_2)_2+2H_2O$ in the process. If n (NO_2) : n (NO) < 1 : 1 in tail gas, it means that there is more nitric oxide than nitrogen dioxide, so the content of NO in tail gas is high, reaction ① is dominant, and the content of $Ca(NO_2)_2$ in the product is higher. When n (NO_2) : n (NO) > 1 : 1, the content of NO_2 is high, and the content of $Ca(NO_3)_2$ in the product increases, so C is incorrect. $Ca(NO_2)_2$ decomposes in acidic solution, and nitrite ion undergoes redox reaction (disproportionation reaction), with the ionic equation $3NO_2^-+2H^+=NO_3^-+2NO\uparrow+H_2O$, so D is correct.)

8. 为吸收工业尾气中的 NO 和 SO_2，同时获得连二亚硫酸钠 ($Na_2S_2O_4$) 和 NH_4NO_3，设计流程如下。下列说法正确的是 (The process is designed to absorb NO and SO_2 in industrial tail gas and obtain sodium dithionite ($Na_2S_2O_4$) and NH_4NO_3 at the same time. Which of the following statements is correct?) ( )

A. 装置 Ⅰ 的作用是吸收 NO 和 SO_2(The function of device Ⅰ is to absorb NO and SO_2)

B. 键角大小：NO_2^-<NO_3^-(Bond angle: NO_2^-<NO_3^-)

C.Ce^{4+}从装置 Ⅲ 的阴极口流出回到装置 Ⅱ 循环使用 (Ce^{4+} flows out from the cathode port of device Ⅲ and returns to device Ⅱ for recycling)

D. 装置 Ⅳ 中氧化 0.1 mol・L^{-1} NO_2^-，至少需要标准状况下 1.12 L O_2(Oxidizing 0.1 mol・L^{-1} NO_2^- in device Ⅳ requires at least 1.12 L O_2 under standard conditions)

答案 (Answer):B

解析 (Analysis)：装置 Ⅰ 中加入 NaOH 溶液吸收 SO_2，装置 Ⅱ 中加入 Ce^{4+}，酸性条件下，NO 和 Ce^{4+}发生氧化还原反应，生成 NO_2^-、NO_3^-，装置 Ⅲ(电解槽) 中阳极发生反应：$Ce^{3+}-e^-=Ce^{4+}$，Ce^{4+}从阳极口流出回到装置 Ⅱ 循环使用，阴极得到 $S_2O_4^{2-}$，装置 Ⅳ 中 NO_2^-被氧气氧化为 NO_3^-，NO_3^-与氨气反应生成硝酸铵。由分析可知，NO 不和 NaOH 反应，装置 Ⅰ 中加入 NaOH 溶液的作用是吸收 SO_2，不能吸收 NO，A 错误；NO_2^-中心原子 N 的价层电子对数为 $2+\frac{1}{2}\times(5+1-2\times2)=3$，$NO_3^-$中心原子 N 的价层电子对数为 $3+\frac{1}{2}\times(5+1-3\times2)=3$，$NO_3^-$和 NO_2^-中心原子 N 都是 sp^2 杂化，NO_3^-中 N 原子上没有孤电子对，NO_2^-中 N 原子上含有 1 个孤电子对，孤电子对越多，键角越小，则键角大小：NO_2^-<NO_3^-，B 正确；装置 Ⅲ(电解槽) 中阳极发生反应：$Ce^{3+}-e^-=Ce^{4+}$，Ce^{4+}从阳极口流出回到装置 Ⅱ 循环使用，C 错误；未说明 0.1 mol・L^{-1} NO_2^-溶液的体积，无法计算需要氧气的物质的量，D 错误。(NaOH solution is added to device Ⅰ to absorb SO_2, and Ce^{4+} is added to device Ⅱ. Under acidic conditions, NO and Ce^{4+} undergo redox reaction to form NO_2^- and NO_3^-. The reaction at the anode of device Ⅲ (electrolytic cell) is: $Ce^{3+}-e^-=Ce^{4+}$, Ce^{4+} flows out from the anode port and returns to device Ⅱ for recycling, and $S_2O_4^{2-}$ is obtained at the cathode. In device Ⅳ, NO_2^- is oxidized to NO_3^- by oxygen, and NO_3^- reacts with ammonia to form ammonium nitrate. As analyzed above, NO does not react with NaOH, and the function of adding NaOH solution to device Ⅰ is to absorb SO_2 but not NO, so A is incorrect. The valence shell electron pair number of central atom N in NO_2^- is $2+\frac{1}{2}\times(5+1-2\times2)=3$, and that in NO_3^- is $3+\frac{1}{2}\times(5+1-3\times2)=3$. The central atom N of both NO_3^- and NO_2^- is sp^2 hybridized. There is no lone pair on N atom in NO_3^-, and 1 lone pair on N atom in NO_2^-. The more lone pairs, the smaller the bond angle, so the bond angle: NO_2^-<NO_3^-, so B is correct. The reaction at the anode of device Ⅲ (electrolytic cell) is: $Ce^{3+}-e^-=Ce^{4+}$, Ce^{4+} flows out from the anode port and returns to device Ⅱ for recycling, so C is incorrect. The volume of 0.1 mol·L⁻¹ NO_2^- solution is not specified, so the amount of oxygen required cannot be calculated, so D is incorrect.)

9.工业上常采用 NaOH 溶液吸收工业尾气中的 NO 和 NO_2，其原理如下。

反应Ⅰ: $NO + NO_2 + 2\,NaOH = 2\,NaNO_2 + H_2O$*

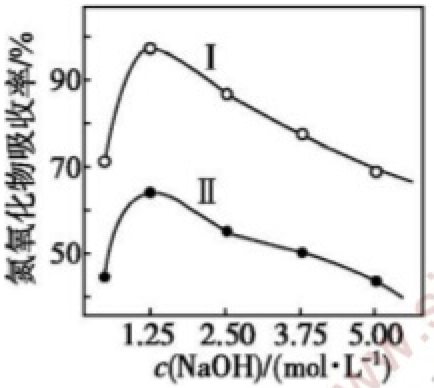
反应Ⅱ: $2\,NO_2 + 2\,NaOH = NaNO_2 + NaNO_3 + H_2O$

NaOH溶液浓度越大黏稠度越高控制其他条件不变，将 NO 和 NO_2体积比为 1∶1、2∶1 的混合气体分别通过体积相同、浓度不同的 NaOH 溶液后，氮氧化物吸收率变化如图所示。下列说法不正确的是 (Industry often uses NaOH solution to absorb NO and NO_2 in industrial tail gas, and the principles are as follows.

Reaction Ⅰ: $NO + NO_2 + 2\,NaOH = 2\,NaNO_2 + H_2O$*

Reaction Ⅱ: $2\,NO_2 + 2\,NaOH = NaNO_2 + NaNO_3 + H_2O$

The higher the concentration of $NaOH$ solution, the greater its viscosity. Keep other conditions unchanged, pass the mixed gases with volume ratio of NO to NO_2 of 1:1 and 2:1 through NaOH solutions with the same volume and different concentrations respectively, and the change of nitrogen oxide absorption rate is shown in the figure. Which of the following statements is incorrect?) ( )



A. 曲线 Ⅰ 表示 NO 和 NO_2体积比为 1 : 1 的混合气体吸收率的变化 (Curve Ⅰ represents the change of absorption rate of mixed gas with volume ratio of NO to NO_2 of 1:1)

B. 当 NaOH 浓度高于 1.25 mol · L⁻¹ 时，吸收率下降的原因可能是 NaOH 溶液黏稠度过高，不利于氮氧化物气体的吸收 (When the concentration of NaOH is higher than 1.25 mol · L⁻¹, the reason for the decrease of absorption rate may be that the viscosity of NaOH solution is too high, which is not conducive to the absorption of nitrogen oxide gas)

C. 向 NO 和 NO_2体积比为 2 : 1 的混合气体中通入少量 O_2，可提高氮氧化物的吸收率 (Passing a small amount of O_2 into the mixed gas with volume ratio of NO to NO_2 of 2:1 can improve the absorption rate of nitrogen oxides)

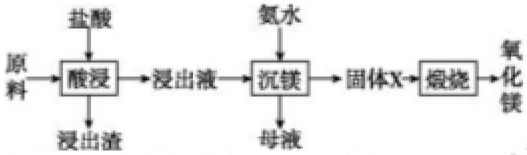
D. 将 1 mol NO 和 3 mol NO_2混合气体通入足量 NaOH 溶液中完全吸收，所得溶液中$\frac{n(NaNO_2)}{n(NaNO_3)}$ =2 : 1(Pass 1 mol NO and 3 mol NO_2 mixed gas into sufficient NaOH solution for complete absorption, and $\frac{n(NaNO_2)}{n(NaNO_3)}$=2 : 1 in the obtained solution)

答案 (Answer):D

解析 (Analysis)：根据反应 Ⅰ 和反应 Ⅱ 可知，NO 和 NO_2体积比为 1 : 1 时吸收率远大于 2 : 1，故曲线 Ⅰ 表示 NO 和 NO_2体积比为 1 : 1 的混合气体吸收率的变化，A 正确；当 NaOH 浓度高于 1.25 mol · L⁻¹ 时，吸收率下降的原因可能是 NaOH 溶液黏稠度过高，不利于氮氧化物气体的吸收，B 正确；向 NO 和 NO_2体积比为 2 : 1 的混合气体中通入少量 O_2，O_2可以将 NO 氧化为 NO_2，可提高氮氧化物的吸收率，C 正确；将 1 mol NO 和 3 mol NO_2混合气体通入足量 NaOH 溶液中完全吸收，由反应 Ⅰ 和 Ⅱ 可知，所得溶液中$\frac{n(NaNO_2)}{n(NaNO_3)}$=3 : 1，D 错误。(According to reaction Ⅰ and reaction Ⅱ, the absorption rate of mixed gas with volume ratio of NO to NO_2 of 1:1 is much higher than that of 2:1, so curve Ⅰ represents the change of absorption rate of mixed gas with volume ratio of NO to

NO_2 of 1:1, so A is correct. When the concentration of NaOH is higher than 1.25 mol · L^{-1}, the reason for the decrease of absorption rate may be that the viscosity of NaOH solution is too high, which is not conducive to the absorption of nitrogen oxide gas, so B is correct. Passing a small amount of O_2 into the mixed gas with volume ratio of NO to NO_2 of 2:1 can oxidize NO to NO_2 and improve the absorption rate of nitrogen oxides, so C is correct. Pass 1 mol NO and 3 mol NO_2 mixed gas into sufficient NaOH solution for complete absorption. According to reaction Ⅰ and Ⅱ, $\frac{n(NaNO_2)}{n(NaNO_3)}$=3 : 1 in the obtained solution, so D is incorrect.)

10.以混有 SiO_2的 $MgCO_3$为原料制备氧化镁的实验流程如图所示。下列说法错误的是 (The experimental process of preparing magnesium oxide from $MgCO_3$ mixed with SiO_2 is shown in the figure. Which of the following statements is incorrect?) ( )



A.“酸浸” 的离子方程式为 $CO_3^{2-}+2H^+=CO_2\uparrow+H_2O$ (The ionic equation of acid leaching is $CO_3^{2-}+2H^+=CO_2\uparrow+H_2O$)

B. 浸出渣的成分是 SiO_2(The component of leaching residue is SiO_2)

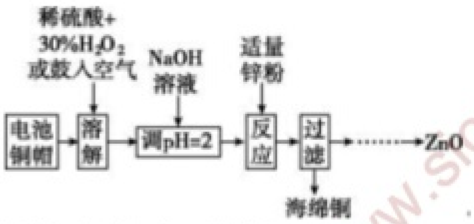
C. 母液的主要溶质是 NH_4Cl (The main solute of mother liquor is NH_4Cl)

D. 固体 X 是 Mg $(OH)_2$(Solid X is Mg $(OH)_2$)

答案 (Answer):A

解析 (Analysis)：$MgCO_3$是微溶物，不可拆成离子，故 A 错误；SiO_2不溶于盐酸，浸出渣为 SiO_2，故 B 正确；加入氨水沉镁的化学方程式为 $MgCl_2+2NH_3 \cdot H_2O=Mg(OH)_2\downarrow+2NH_4Cl$，母液的主要溶质是 NH_4Cl，固体 X 是 Mg $(OH)_2$，故 C、D 正确。($MgCO_3$ is a slightly soluble substance and cannot be dissociated into ions, so A is incorrect. SiO_2 is insoluble in hydrochloric acid, and the leaching residue is SiO_2, so B is correct. The chemical equation for magnesium precipitation by adding ammonia water is $MgCl_2+2NH_3 \cdot H_2O=Mg(OH)_2\downarrow+2NH_4Cl$. The main solute of mother liquor is NH_4Cl, and solid X is Mg $(OH)_2$, so C and D are correct.)

11.为研究废旧电池的再利用，实验室利用废旧电池的铜帽 (主要成分为 Zn 和 Cu) 回收 Cu 并制备 ZnO 的部分实验过程如图所示。下列叙述错误的是 (To study the reuse of waste batteries, the laboratory uses the copper caps of waste batteries (main components are Zn and Cu) to recover Cu and prepare ZnO. The partial experimental process is shown in the figure. Which of the following statements is incorrect?) ( )



A.“溶解” 操作中溶液温度不宜过高 (The solution temperature should not be too high in the dissolution operation)

B. 铜帽溶解后，将溶液加热至沸腾以除去溶液中过量的氧气或 H_2O_2(After the copper cap is dissolved, heat the solution to boiling to remove excess oxygen or H_2O_2 in the solution)

C. 与锌粉反应的离子可能为 Cu^{2+}、H^+(The ions reacting with zinc powder may be Cu^{2+} and H^+)

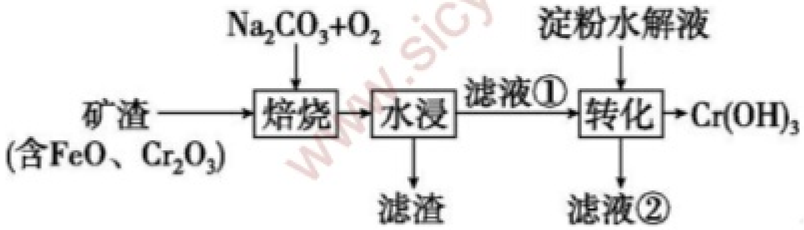
D.“过滤” 操作后，将滤液蒸发结晶、过滤、洗涤、干燥后，高温灼烧即可得纯净的 ZnO (After the filtration operation, evaporate and crystallize the filtrate, filter, wash and dry, then burn at high temperature to obtain pure ZnO)

答案 (Answer):D

解析 (Analysis)：“溶解” 操作中溶液温度不宜过高，否则过氧化氢会分解，故 A 正确；铜帽溶解后，溶液中存在过氧化氢和氧气，将溶液加热至沸腾以除去溶液中过量的氧气或 H_2O_2，防止后面消耗过多的锌粉，故 B 正确；固体溶解后溶液中存在 Cu^{2+}、H^+，都可以与锌粉反应，故 C 正确；“过滤” 操

作后滤液中含有硫酸锌和硫酸钠，将滤液蒸发结晶、过滤、洗涤、干燥后，高温灼烧不能得到纯净的 ZnO，故 D 错误。(The solution temperature should not be too high in the dissolution operation, otherwise hydrogen peroxide will decompose, so A is correct. After the copper cap is dissolved, hydrogen peroxide and oxygen exist in the solution. Heat the solution to boiling to remove excess oxygen or H_2O_2 in the solution to prevent excessive consumption of zinc powder later, so B is correct. Cu^{2+} and H^+ exist in the solution after solid dissolution, both of which can react with zinc powder, so C is correct. The filtrate after filtration contains zinc sulfate and sodium sulfate. After evaporating and crystallizing the filtrate, filtering, washing and drying, burning at high temperature cannot obtain pure ZnO, so D is incorrect.)

12.某工厂采用如下工艺制备 $Cr(OH)_3$，已知焙烧后 Cr 元素以 + 6 价形式存在，下列说法错误的是 (A factory uses the following process to prepare $Cr(OH)_3$. It is known that Cr element exists in +6 valence after roasting. Which of the following statements is incorrect?) ( )



A.“焙烧” 中产生 CO_2(CO₂ is produced in roasting)

B. 滤渣的主要成分为 $Fe(OH)_2$(The main component of filter residue is $Fe(OH)_2$)

C. 滤液①中 Cr 元素的主要存在形式为 CrO_4^{2-}(The main existing form of Cr element in filtrate ① is CrO_4^{2-})

D. 淀粉水解液中的葡萄糖起还原作用 (Glucose in starch hydrolysate acts as a reducing agent)

答案 (Answer):B

解析 (Analysis)：由题意可知，“焙烧” 过程中 FeO、Cr_2O_3均被 O_2氧化，同时 Cr 转化为对应钠盐，“水浸” 时 Fe_2O_3不溶，过滤除去滤渣，滤液①中存在 Na_2CrO_4，与淀粉的水解产物葡萄糖发生氧化还原反应得到氢氧化铬沉淀。“焙烧” 过程中 Cr_2O_3发生反应产生 CO_2：$2Cr_2O_3+3O_2+4Na_2CO_3 \overset{高温}{=} 4Na_2CrO_4+4CO_2$，A 正确；“焙烧” 过程中 FeO 被氧化，滤渣的主要成分为 Fe_2O_3，B 错误；滤液①中 Cr 元素的化合价是 + 6 价，溶液显碱性，所以 Cr 元素主要存在形式为 CrO_4^{2-}，C 正确；“转化” 过程中 Cr 元素由 + 6 价降低为 + 3 价，CrO_4^{2-}发生还原反应，淀粉水解液中的葡萄糖起还原作用，D 正确。(As known from the question, FeO and Cr_2O_3 are both oxidized by O_2 during roasting, and Cr is converted into the corresponding sodium salt at the same time. Fe_2O_3 is insoluble during water leaching, and the filter residue is removed by filtration. Na_2CrO_4 exists in filtrate ①, which undergoes redox reaction with glucose (hydrolysate of starch) to obtain chromium hydroxide precipitate. The reaction of Cr_2O_3 during roasting produces CO_2: $2Cr_2O_3+3O_2+4Na_2CO_3 \overset{hightemperature}{=} 4Na_2CrO_4+4CO_2$, so A is correct. FeO is oxidized during roasting, and the main component of filter residue is Fe_2O_3, so B is incorrect. The valence of Cr element in filtrate ① is +6, and the solution is alkaline, so the main existing form of Cr element is CrO_4^{2-}, so C is correct. During transformation, the valence of Cr element decreases from +6 to +3, CrO_4^{2-} undergoes reduction reaction, and glucose in starch hydrolysate acts as a reducing agent, so D is correct.)

13.碳酸亚铁 ($FeCO_3$) 能够帮助治疗缺铁性贫血，属于一种补铁的营养补充剂。制备 $FeCO_3$的工艺流程如图所示。已知：“还原” 工序中不生成 S 单质。下列说法错误的是 (Ferrous carbonate ($FeCO_3$) can help treat iron deficiency anemia and is an iron-supplementing nutritional supplement. The process flow for preparing $FeCO_3$ is shown in the figure. Known: no elemental S is generated in the reductionprocess. Which of the following statements is incorrect?) ( )

A.FeS_2中 S 的化合价为 - 1 价 (The valence of S in FeS_2 is -1)
B."酸浸" 时，物质 X 为 Fe_3O_4(During acid leaching, substance X is Fe_3O_4)
C."还原" 时发生反应的离子方程式为 $14Fe^{3+}+FeS_2+8H_2O=15Fe^{2+}+2SO_4^{2-}+16H^+$(The ionic equation of the reaction during reduction is $14Fe^{3+}+FeS_2+8H_2O=15Fe^{2+}+2SO_4^{2-}+16H^+$)
D."沉铁" 时，HCO_3^-的电离平衡正向移动 (During iron precipitation, the ionization equilibrium of HCO_3^- moves forward)

答案 (Answer):B

解析 (Analysis)：该工艺流程原料为 FeS_2，产品为 $FeCO_3$，FeS_2在空气中焙烧发生反应为 $4FeS_2+11O_2\overset{高温}{=}2Fe_2O_3+8SO_2$，物质 X 为 Fe_2O_3，酸浸得 $Fe_2(SO_4)_3$溶液，与 FeS_2在 "还原" 工艺发生反应得到 Fe^{2+}，通过加 NH_4HCO_3沉铁得到 $FeCO_3$。FeS_2中 Fe 的化合价 + 2 价，S 的化合价为 - 1 价，A 正确；FeS_2在空气中焙烧发生反应为 $4FeS_2+11O_2\overset{高温}{=}2Fe_2O_3+8SO_2$，物质 X 为 Fe_2O_3，B 错误；酸浸得 $Fe_2(SO_4)_3$溶液，与 FeS_2在 "还原" 工序发生反应得到 Fe^{2+}，离子方程式为 $14Fe^{3+}+FeS_2+8H_2O=15Fe^{2+}+2SO_4^{2-}+16H^+$，C 正确；"沉铁" 时反应为 $Fe^{2+}+2HCO_3^-=FeCO_3\downarrow+H_2O+CO_2\uparrow$，$Fe^{2+}$与 HCO_3^-电离产生的 CO_3^{2-}结合生成难溶物 $FeCO_3$，使 $HCO_3^-\rightleftharpoons H^++CO_3^{2-}$平衡正移，D 正确。(The raw material of this process is FeS_2, and the product is $FeCO_3$. The reaction of FeS_2 roasting in air is $4FeS_2+11O_2\overset{hightemperature}{=}2Fe_2O_3+8SO_2$. Substance X is Fe_2O_3, which is leached by acid to obtain $Fe_2(SO_4)_3$ solution. It reacts with FeS_2 in the reduction process to obtain Fe^{2+}, and $FeCO_3$ is obtained by adding NH_4HCO_3 for iron precipitation. The valence of Fe in FeS_2 is +2, and the valence of S is -1, so A is correct. The reaction of FeS_2 roasting in air is $4FeS_2+11O_2\overset{hightemperature}{=}2Fe_2O_3+8SO_2$, and substance X is Fe_2O_3, so B is incorrect. $Fe_2(SO_4)_3$ solution is obtained by acid leaching, which reacts with FeS_2 in the reduction process to obtain Fe^{2+}, with the ionic equation $14Fe^{3+}+FeS_2+8H_2O=15Fe^{2+}+2SO_4^{2-}+16H^+$, so C is correct. The reaction during iron precipitation is $Fe^{2+}+2HCO_3^-=FeCO_3\downarrow+H_2O+CO_2\uparrow$. Fe^{2+} combines with CO_3^{2-} produced by ionization of HCO_3^- to form insoluble $FeCO_3$, making the equilibrium $HCO_3^-\rightleftharpoons H^++CO_3^{2-}$ move forward, so D is correct.)

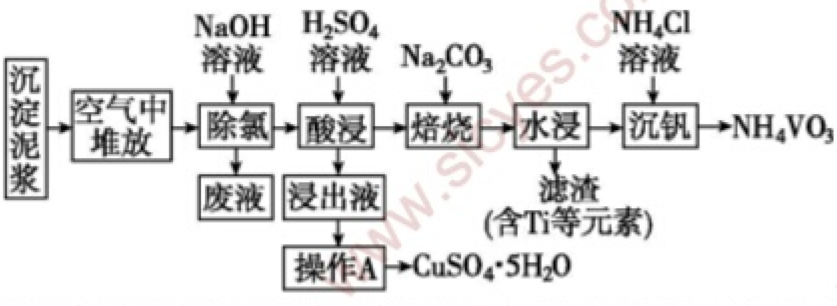
14.某种以辉铜矿 [Cu_2S，含 Fe_3O_4、SiO_2杂质] 为原料制备 [Cu (NH_3)₄] $SO_4\cdot H_2O$ 的流程如图所示，下列说法错误的是 ( )(A process for preparing [Cu (NH_3)$_4$] $SO_4\cdot H_2O$ from chalcocite [Cu_2S, containing Fe_3O_4 and SiO_2 impurities] is shown in the figure. Which of the following statements is incorrect?)

A. 向溶液 D 中加入无水乙醇并用玻璃棒摩擦试管壁可析出目标产物（Adding anhydrous ethanol to solution D and rubbing the test tube wall with a glass rod can precipitate the target product.)
B. 向溶液 C 中加入硫氰化钾溶液后呈现红色（Adding potassium thiocyanate solution to solution C results in a red color.)
C. 煅烧过程中涉及的主要方程式为 $2Cu_2S+5O_2\overset{高温}{=}4CuO+2SO_3$（The main equation involved in the calcination process is $2Cu_2S + 5O_2\overset{hightemperature}{=}4CuO + 2SO_3$.)
D. $[Cu(NH_3)_4]SO_4\cdot H_2O$ 中配位原子为 N（The coordination atom in $[Cu(NH_3)_4]SO_4\cdot H_2O$ is N.)

答案（Answer）：C

解析（Analysis）：向辉铜矿中通入氧气进行煅烧，发生 $Cu_2S+2O_2 \overset{高温}{=} 2CuO+SO_2$，固体 B 为 CuO、Fe_3O_4和 SiO_2，加入硫酸过滤后得到含有 Cu^{2+}、Fe^{3+}、Fe^{2+}的溶液，加入过量氨水，得到氢氧化铁沉淀、氢氧化亚铁沉淀和 $[Cu(NH_3)_4]SO_4$溶液，再经过处理得到 $[Cu(NH_3)_4]SO_4 \cdot H_2O$。溶液 D 为 $[Cu(NH_3)_4]SO_4$溶液，向溶液 D 中加入无水乙醇，可减小溶液的极性、降低产物的溶解性，用玻璃棒摩擦试管壁可析出目标产物，A 正确；溶液 C 中溶质的主要成分为 $FeSO_4$、$CuSO_4$和 $Fe_2(SO_4)_3$，加入硫氰化钾溶液会产生红色，B 正确；辉铜矿通入氧气进行煅烧，发生反应 $Cu_2S+2O_2 \overset{高温}{=} 2CuO+SO_2$，C 错误；$[Cu(NH_3)_4]SO_4 \cdot H_2O$ 中配体是氨分子，配位原子为 N，D 正确。（Chalcocite is calcined by introducing oxygen, and the reaction $Cu_2S + 2O_2 \overset{hightemperature}{=} 2CuO + SO_2$ occurs. Solid B is CuO, Fe_3O_4 and SiO_2. After adding sulfuric acid and filtering, a solution containing Cu^{2+}, Fe^{3+} and Fe^{2+} is obtained. Adding excess ammonia water yields iron hydroxide precipitate, ferrous hydroxide precipitate and $[Cu(NH_3)_4]SO_4$ solution, which is further processed to obtain $[Cu(NH_3)_4]SO_4 \cdot H_2O$. Solution D is $[Cu(NH_3)_4]SO_4$ solution. Adding anhydrous ethanol to solution D can reduce the polarity of the solution and the solubility of the product, and rubbing the test tube wall with a glass rod can precipitate the target product, so A is correct. The main solutes in solution C are $FeSO_4$, $CuSO_4$ and $Fe_2(SO_4)_3$, and adding potassium thiocyanate solution produces a red color, so B is correct. Chalcocite is calcined by introducing oxygen, and the reaction $Cu_2S + 2O_2 \overset{hightemperature}{=} 2CuO + SO_2$ occurs, so C is incorrect. The ligand in $[Cu(NH_3)_4]SO_4 \cdot H_2O$ is ammonia molecule, and the coordination atom is N, so D is correct.)

15.某工厂废液经简易处理后的沉淀泥浆中含有大量废铜丝和少量 V、Ti、Fe、Si 元素氯化物及复杂化合物，一种以该沉淀泥浆为原料回收铜与钒的工艺流程如图所示。已知:常温下 $K_{sp}[Fe(OH)_3]=2.79\times10^{-39}$、$K_{sp}[Cu(OH)_2]=2.2\times10^{-20}$,溶液中的离子浓度小于 $10^{-5}\ mol{\cdot}L^{-1}$时,可以认为已经被除尽。下列说法错误的是 ( )(The precipitation slurry after simple treatment of a factory's waste liquid contains a large amount of waste copper wires, a small amount of chlorides and complex compounds of V, Ti, Fe and Si elements. A process flow for recovering copper and vanadium from the precipitation slurry is shown in the figure. Known: At room temperature, $K_{sp}[Fe(OH)_3]=2.79\times10^{-39}$ and $K_{sp}[Cu(OH)_2]=2.2\times10^{-20}$. When the ion concentration in the solution is less than $10^{-5}\ mol{\cdot}L^{-1}$, it can be considered completely removed.Which of the following statements is incorrect?)



A. 泥浆在空气中堆放后由灰色变为疏松的绿色粉状，是因为铜被氧化，便于后续酸浸时铜进入溶液（The slurry turns from gray to loose green powder after being stacked in air because copper is oxidized, which facilitates copper to enter the solution during subsequent acid leaching.）

B. 流程中利用 $Cu_2(OH)_3Cl+OH^-=2Cu(OH)_2+Cl^-$除氯，可以减少酸浸时含氯化合物的挥发（Chlorine is removed by the reaction $Cu_2(OH)_3Cl + OH^- = 2Cu(OH)_2 + Cl^-$ in the process, which can reduce the volatilization of chlorine-containing compounds during acid leaching.）

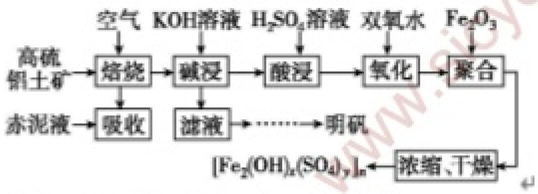
C. 若浸出液中 Cu^{2+}的浓度为 0.1 mol·L⁻¹，则酸浸时可调 pH=5（If the concentration of Cu^{2+} in the leachate is 0.1 mol·L⁻¹, the pH can be adjusted to 5 during acid leaching.）

D. 焙烧加入 Na_2CO_3的目的是使 V 元素转化成可溶性的钒酸盐（The purpose of adding Na_2CO_3 during calcination is to convert element V into soluble vanadate.）

答案（Answer）：C

解析（Analysis）：由泥浆堆放后由灰色变为疏松的绿色粉状可以得出是铜被氧化，酸浸时可以进入溶液便于后期制备胆矾，A 正确；常温下除氯，Cl^-进入废液，可以在酸浸时减少 HCl 挥发，更加环保，B 正确；pH=5 时，$c(OH^-)=10^{-9}\ mol\cdot L^{-1}$，经计算 $c(Fe^{3+})=2.79\times10^{-12}\ mol\cdot L^{-1}$，能使 Fe^{3+}除尽，$c(Cu^{2+})=2.2\times10^{-2}\ mol\cdot L^{-1}$<0.1 $mol\cdot L^{-1}$，Cu^{2+}也会随之沉淀，C 错误；焙烧时加入 Na_2CO_3，可以使 V 元素转化成可溶性的钒酸盐，水浸时进入溶液，水浸除掉杂质元素，更好地回收钒，D 正确。(The slurry turns from gray to loose green powder after stacking because copper is oxidized, which can enter the solution during acid leaching to facilitate the later preparation of copper sulfate pentahydrate, so A is correct. Chlorine is removed at room temperature and Cl^- enters the waste liquid, which can reduce the volatilization of HCl during acid leaching and is more environmentally friendly, so B is correct. When pH=5, $c(OH^-)=10^{-9}\ mol\cdot L^{-1}$. Calculation shows that $c(Fe^{3+})=2.79\times10^{-12}\ mol\cdot L^{-1}$, which can completely remove Fe^{3+}; $c(Cu^{2+})=2.2\times10^{-2}\ mol\cdot L^{-1}$<0.1 $mol\cdot L^{-1}$, so Cu^{2+} will also precipitate, so C is incorrect. Adding Na_2CO_3 during calcination can convert element V into soluble vanadate, which enters the solution during water leaching. Water leaching removes impurity elements to better recover vanadium, so D is correct.)

16.以高硫铝土矿 (主要成分为 Fe_2O_3、Al_2O_3、SiO_2，还含有少量 FeS_2和硫酸盐) 为原料制备聚合硫酸铁 {$[Fe_2(OH)_x(SO_4)_y]_n$} 和明矾的部分工艺流程如图所示。已知：赤泥液的主要成分为 Na_2CO_3。下列说法错误的是 ( )(A partial process flow for preparing polymeric ferric sulfate {$[Fe_2(OH)_x(SO_4)_y]_n$} and alum from high-sulfur bauxite (main components are Fe_2O_3, Al_2O_3, SiO_2, also containing a small amount of FeS_2 and sulfate) is shown in the figure. Known: the main component of red mud liquid is Na_2CO_3. Which of the following statements is incorrect?)



A. 赤泥液的作用是吸收 “焙烧” 阶段中产生的 SO_2（The function of red mud liquid is to absorb SO_2 produced in the calcination stage.)

B. 从 “滤液” 到 “明矾” 的过程中还应有 “除硅” 步骤（There should also be a silicon removal step from the filtrate to alum)

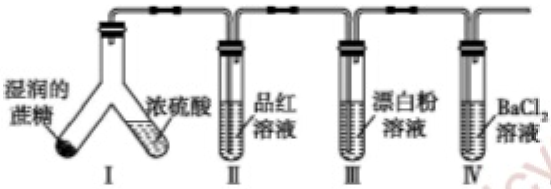
C. 聚合硫酸铁可用于净化自来水，与其组成中的 Fe^{3+}具有氧化性有关（Polymeric ferric sulfate can be used to purify tap water, which is related to the oxidizing property of Fe^{3+} in its composition.)

D. 在 “聚合” 阶段，若增加 Fe_2O_3用量，会使 $[Fe_2(OH)_x(SO_4)_y]_n$中 x 变大（In the polymerization stage, increasing the amount of Fe_2O_3 will increase x in $[Fe_2(OH)_x(SO_4)_y]_n$.)

答案（Answer）：C

解析（Analysis）：焙烧时，FeS_2与 O_2反应生成 Fe_2O_3和 SO_2，反应方程式为 $4FeS_2+11O_2 \overset{高温}{=} 2Fe_2O_3+8SO_2$，用赤泥液中的 Na_2CO_3吸收生成的 SO_2，A 正确；由于 “碱浸” 造成二氧化硅溶于氢氧化钾溶液，滤液中含硅酸盐，从 “滤液” 到 “明矾” 的过程中，需要进行 “除硅” 步骤，B 正确；聚合硫酸铁可用于净化自来水，与其组成中的 Fe^{3+}具有水解产生胶体的性质有关，与其组成中的 Fe^{3+}具有氧化性无关，C 错误；在 “聚合” 阶段，若增加 Fe_2O_3用量，碱性增强，x 变大，D 正确。(During calcination, FeS_2 reacts with O_2 to form Fe_2O_3 and SO_2, and the reaction equation is $4FeS_2 + 11O_2 \overset{hightemperature}{=} 2Fe_2O_3 + 8SO_2$. Na_2CO_3 in red mud liquid is used to absorb the generated SO_2, so A is correct. Since silicon dioxide dissolves in potassium hydroxide solution during alkali leaching, the filtrate contains silicate. A silicon removal step is required from the filtrate to alum, so B is correct. Polymeric ferric sulfate can be used to purify tap water because Fe^{3+} in its composition hydrolyzes to form colloid, which has nothing to do with the oxidizing property of Fe^{3+}, so C is incorrect. In the polymerizations tage, increasing the amount of Fe_2O_3 enhances the alkalinity and increases x, so D is correct.)

17.某化学兴趣小组为探究蔗糖与浓硫酸的反应设计了如图所示实验装置。向左倾斜 Y 形试管使反应发生，下列说法正确的是 ( )(A chemistry interest group designed the experimental device shown in the figure to explore the reaction between sucrose and concentrated sulfuric acid. Tilt the Y-shaped test tube to the left to initiate the reaction. Which of the following statements is correct?)

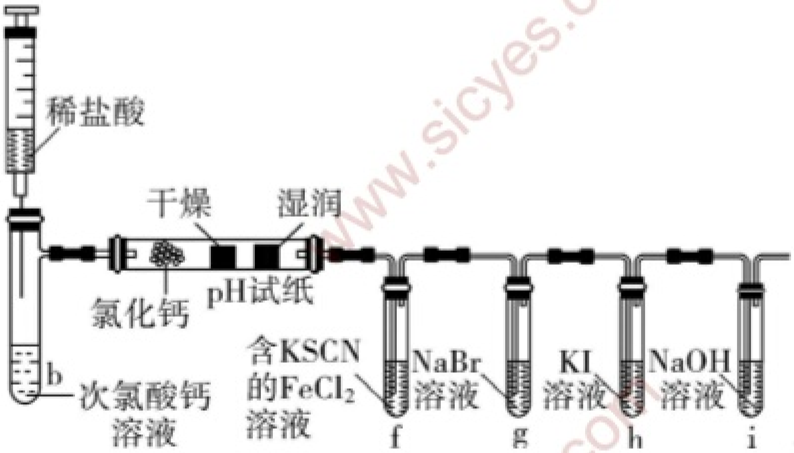


A. 装置 Ⅱ 中品红溶液褪色，证明 SO_2具有氧化性（The fuchsine solution fades in device Ⅱ, proving that SO_2 has oxidizing property.)
B. 装置 Ⅲ 中产生白色沉淀，其主要成分为 $CaCO_3$和 $CaSO_4$（White precipitate is formed in device Ⅲ, and its main components are $CaCO_3$ and $CaSO_4$.)
C. 装置 Ⅳ 中无明显现象（There is no obvious phenomenon in device Ⅳ.)
D. 装置不变，仅将装置 Ⅰ 中的蔗糖换成木炭，也能出现相同的现象（Keep the device unchanged and only replace sucrose in device Ⅰ with charcoal, the same phenomenon will occur.)

答案（Answer）：C

解析（Analysis）：蔗糖与浓硫酸反应产生 CO_2和 SO_2，SO_2具有漂白性，使品红溶液褪色，A 错误；漂白粉的主要成分是 $CaCl_2$和 $Ca(ClO)_2$，$Ca(ClO)_2$具有氧化性，通入 CO_2和 SO_2后反应生成 $CaCO_3$和 $CaSO_4$沉淀，B 错误；CO_2和 SO_2均不能与 $BaCl_2$反应，故装置 Ⅳ 无明显现象，C 正确；木炭与浓硫酸反应需要加热，仅将装置 Ⅰ 中的蔗糖换成木炭，不能出现相同的现象，D 错误。（Sucrose reacts with concentrated sulfuric acid to produce CO_2 and SO_2. SO_2 has bleaching property and fades the fuchsine solution, so A is incorrect. The main components of bleaching powder are $CaCl_2$ and $Ca(ClO)_2$. $Ca(ClO)_2$ has oxidizing property and reacts with CO_2 and SO_2 to form $CaCO_3$ and $CaSO_4$ precipitates, so B is incorrect. Neither CO_2 nor SO_2 can react with $BaCl_2$, so there is no obvious phenomenon in device Ⅳ, so C is correct. The reaction between charcoal and concentrated sulfuric acid requires heating. Only replacing sucrose with charcoal in device Ⅰ cannot produce the same phenomenon, so D is incorrect.)

18.化学兴趣小组采用次氯酸钙与稀盐酸反应制取氯气，并探究了氯气的性质。实验装置如图所示，下列说法正确的是 ( )(A chemistry interest group prepared chlorine gas by reacting calcium hypochlorite with dilute hydrochloric acid and explored the properties of chlorine gas. The experimental device is shown in the figure. Which of the following statements is correct?)



A. 湿润的 pH 试纸先变红后褪色，说明 Cl_2有酸性和漂白性（The wet pH test paper turns red first and then fades, indicating that the solution formed by Cl_2 dissolving in water has acidity and bleaching property.）

B. f 处溶液变红是因为 Fe^{2+}被氧化为 Fe^{3+}，Fe^{3+}遇 KSCN 生成弱电解质 $Fe(SCN)_3$（The solution turns red at f because Fe^{2+} is oxidized to Fe^{3+}, and Fe^{3+} reacts with KSCN to form weak electrolyte $Fe(SCN)_3$.）

C. g 处变为橙色，h 处变为黄色，说明元素非金属性：Cl > Br > I（g turns orange and h turns yellow, indicating the non-metallic property of elements: Cl > Br > I.）

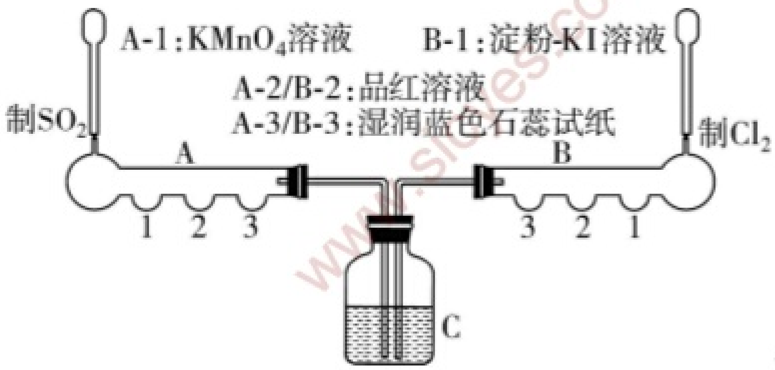
D. 等量 Cl_2分别单独缓慢通过 g、h、i 试管时，生成的氧化产物的物质的量之比为 1 : 1 : 1（When equal amounts of Cl_2 slowly pass through test tubes g, h and i separately, the molar ratio of the generated oxidation products is 1:1:1.）

答案（Answer）：D

解析（Analysis）：次氯酸钙溶液与稀盐酸反应可产生氯气，氯气中混杂的水蒸气可被氯化钙吸收；氯气无漂白性，不能使干燥的 pH 试纸褪色；氯气与水发生反应：$Cl_2+H_2O=HCl+HClO$，其中 HCl 水溶液显酸性，HClO 有漂白性，因此氯气能使湿润的 pH 试纸先变红后褪色；氯气具有强氧化性，可通过 f、g、h 实验验证。NaOH 溶液吸收尾气，防止污染；据此解答。根据分析可知，湿润的 pH 试纸先变红后褪色，说明氯气溶于水后的溶液具有酸性和漂白性，A 错误；已知 Cl_2可将 Fe^{2+}氧化为 Fe^{3+}，Fe^{3+}遇 KSCN 溶液生成红色络合物 $Fe(SCN)_3$，f 处现象为溶液变红、但不出现红色沉淀，B 错误；在试管 g、h 中分别发生氧化还原反应 $Cl_2+2Br^-=Br_2+2Cl^-$，$Cl_2+2I^-=I_2+2Cl^-$，其中氧化性：$Cl_2 > Br_2$，$Cl_2 > I_2$，则元素的非金属性 Cl > Br，Cl > I，该实验不能证明 Br 和 I 元素的非金属性强弱，C 错误；等量 Cl_2分别单独缓慢通过试管 g、h、i 时，分别发生的氧化还原反应为 $Cl_2+2NaBr=Br_2+2NaCl$、$Cl_2+2KI=I_2+2KCl$、$Cl_2+2NaOH=NaCl+NaClO+H_2O$，其中氧化产物分别为 Br_2、I_2、NaClO，根据化学方程式可计算出等物质的量的氯气参与反应，在 g、h、i 试管中生成的氧化产物物质的量之比为 1 : 1 : 1，D 正确。（Calcium hypochlorite solution reacts with dilute hydrochloric acid to produce chlorine gas, and the water vapor mixed in chlorine gas can be absorbed by calcium chloride; chlorine gas has no bleaching property and cannot fade the dry pH test paper; chlorine gas reacts with water: $Cl_2 + H_2O = HCl + HClO$, in which the aqueous solution of HCl is acidic and HClO has bleaching property, so chlorine gas can make the wet pH test paper turn red first and then fade; chlorine gas has strong oxidizing property, which can be verified by experiments f, g and h. NaOH solution absorbs tail gas to prevent pollution; answer accordingly. According to the analysis, the wet pH test paper turns red first and then fades, indicating that the solution after chlorine gas dissolves in water has acidity and bleaching property, so A is incorrect. It is known that Cl_2 can oxidize Fe^{2+} to Fe^{3+}, and Fe^{3+} reacts with KSCN solution to form red complex $Fe(SCN)_3$. The phenomenon at f is that the solution turns red without red precipitate, so B is incorrect. Redox reactions occur in test tubes g and h respectively: $Cl_2 + 2Br^- = Br_2 + 2Cl^-$, $Cl_2 + 2I^- = I_2 + 2Cl^-$, in which oxidizing property: $Cl_2 > Br_2$, $Cl_2 > I_2$, so the non-metallic property of elements is Cl > Br and Cl > I. This experiment cannot prove the non-metallic property of Br and I, so C is incorrect. When equal amounts of Cl_2 slowly pass through test tubes g, h and i separately, the redox reactions are

$Cl_2 + 2NaBr = Br_2 + 2NaCl$, $Cl_2 + 2KI = I_2 + 2KCl$, $Cl_2 + 2NaOH = NaCl + NaClO + H_2O$, in which the oxidation products are Br_2, I_2 and NaClO respectively. According to the chemical equations, the molar ratio of oxidation products generated in test tubes g, h and i is 1:1:1 when equal moles of chlorine gas participate in the reaction, so D is correct.)

19.SO_2和 Cl_2两种气体的制备和性质一体化探究实验装置 (已省略夹持装置) 如图所示。下列说法正确的是 ( )
(The integrated experimental device for the preparation and property exploration of SO_2 and Cl_2 gases (clamping devices omitted) is shown in the figure. Which of the following statements is correct?)

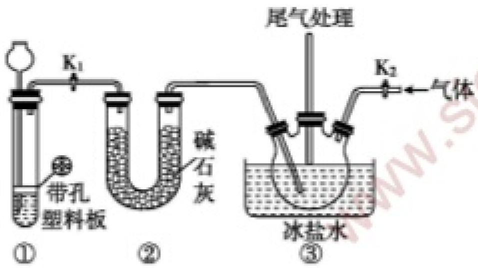


A. 制备氯气可以采用浓盐酸与 MnO_2反应（Chlorine gas can be prepared by reacting concentrated hydrochloric acid with MnO_2 under heating.)
B. A-2/B-2 两处的品红溶液变色的原理相同，都是利用其氧化性（The fuchsine solutions at A-2/B-2 change color by different principles.)
C. A-3/B-3 两处的实验现象是一样的，都是蓝色石蕊试纸变红（The experimental phenomena at A-3/B-3 are different.)
D. 试剂瓶 C 中的溶液若是氯化钡溶液，则会出现白色沉淀（If the solution in reagent bottle C is barium chloride solution, white precipitate will appear.)

答案（Answer）：D

解析（Analysis）：左侧装置制备和检验 SO_2的性质，A-1 为酸性高锰酸钾溶液，可以检验二氧化硫的还原性，A-2 为品红溶液，可以检验二氧化硫的漂白性，A-3 为湿润蓝色石蕊试纸，可以检验二氧化硫溶于水后溶液的酸碱性；右侧装置制备和检验 Cl_2的性质，B-1 为淀粉 - KI 溶液，可以检验 Cl_2的氧化性，B-2 为品红溶液，可以检验 Cl_2与水反应生成的 HClO 的氧化性，B-3 为湿润蓝色石蕊试纸，可以检验 Cl_2溶于水后溶液的酸碱性及氧化性。浓盐酸与 MnO_2需要加热才可以产生氯气，该装置无法加热，A 错误；A-2 品红溶液褪色是由于二氧化硫的漂白性，B-2 品红溶液褪色是由于 Cl_2与水反应生成的 HClO 具有氧化性，B 错误；二氧化硫溶于水后溶液为酸性，使蓝色石蕊试纸变红，Cl_2溶于水后产生盐酸和次氯酸，能使蓝色石蕊试纸先变红后褪色，C 错误；试剂瓶 C 中，Cl_2能将 SO_2氧化为硫酸，与氯化钡溶液发生反应，产生硫酸钡白色沉淀，D 正确。（The left device prepares and tests the properties of SO_2. A-1 is acid potassium permanganate solution, which can test the reducibility of sulfur dioxide; A-2 is fuchsine solution, which can test the bleaching property of sulfur dioxide; A-3 is wet blue litmus test paper, which can test the acidity and alkalinity of the solution after sulfur dioxide dissolves in water. The right device prepares and tests the properties of Cl_2. B-1 is starch-KI solution, which can test the oxidizing property of Cl_2; B-2 is fuchsine solution, which can test the oxidizing property of HClO generated by the reaction of Cl_2 and water; B-3 is wet blue litmus test paper, which can test the acidity, alkalinity and oxidizing property of the solution after Cl_2 dissolves in water. Concentrated hydrochloric acid and MnO_2 need heating to produce chlorine gas, and this device cannot be heated, so A is incorrect. The fuchsine solution at A-2 fades due to the bleaching property of sulfur dioxide, and the fuchsine solution at B-2 fades due to the oxidizing property of HClO generated by the reaction of Cl_2 and water, so B is incorrect. The solution after sulfur dioxide dissolves in water is acidic, turning the blue litmus test paper red; Cl_2 dissolves in water to produce hydrochloric acid and hypochlorous acid, which can make the blue litmus test paper turn red first and then fade, so C is incorrect. In reagent bottle C, Cl_2 can oxidize SO_2 to sulfuric acid, which reacts with barium chloride solution to produce white barium sulfate precipitate, so D is correct.)

20.常温下，氧氯化氮 (NOCl，熔点：-64.5 ℃；沸点：-5.5 ℃) 是一种遇潮气水解的气体，在实验中可利用一氧化氮和氯气反应制备，其装置如图所示。下列说法中正确的是 ( )(At room temperature, nitrosyl chloride (NOCl, melting point: -64.5 ℃; boiling point: -5.5 ℃) is a gas that hydrolyzes in humid air. It can be prepared by the reaction of nitric oxide and chlorine gas in the experiment, and its device is shown in the figure. Which of the following statements is correct?)

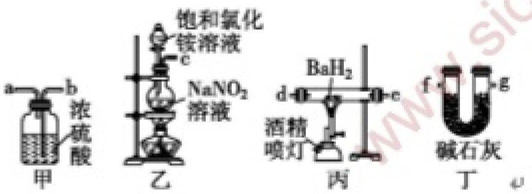


A. 氧氯化氮 (NOCl) 遇到潮湿的空气会出现白烟（Nitrosyl chloride (NOCl) produces white mist when exposed to humid air.)
B. 装置①为浓盐酸和 $KMnO_4$反应，制备所需的 Cl_2（Device ① prepares NO by reacting suitable reagents.)
C. 装置①可用于 Na_2O_2和 H_2O 反应制备 O_2的实验，便于控制反应的发生与停止（Device ① cannot be used to prepare O_2 by reacting Na_2O_2 with H_2O.)
D. 装置③中的冰盐水可使氧氯化氮冷凝为液体，减少挥发（The ice brine in device ③ can condense nitrosyl chloride into liquid and reduce volatilization.)

答案（Answer）：D
解析（Analysis）：装置①中产生的气体能用碱石灰干燥，则生成的气体为 NO，装置③中从右侧通入的气体为 Cl_2。氧氯化氮中的氯为 - 1 价，因此水解过程会产生氯化氢，从而出现白雾，而不是白烟，A 错误；装置①为简易气体发生装置，要求反应物之一必须为块状固体，而 $KMnO_4$为粉末状固体，且能溶于水，达不到控制反应停止的目的，且生成的 Cl_2能被碱石灰吸收，故①中制取的气体是 NO，B 错误；Na_2O_2为粉末状固体，易从塑料板的小孔进入溶液中，无法控制反应的停止，C 错误；冰盐水的温度低，能将氧氯化氮冷却，并由气体转变为液体，以便收集于三颈烧瓶中，D 正确。（The gas produced in device ① can be dried by soda lime, so the generated gas is NO, and the gas introduced from the right side in device ③ is Cl_2. Chlorine in nitrosyl chloride is -1 valence, so hydrogen chloride is produced during hydrolysis, resulting in white mist instead of white smoke, so A is incorrect. Device ① is a simple gas generator, which requires one of the reactants to be a block solid. $KMnO_4$ is a powder solid and soluble in water, which cannot control the stop of the reaction, and the generated Cl_2 can be absorbed by soda lime, so the gas prepared in ① is NO, so B is incorrect. Na_2O_2 is a powder solid, which easily enters the solution through the small holes of the plastic plate and cannot control the stop of the reaction, so C is incorrect. The temperature of ice brine is low, which can cool nitrosyl chloride from gas to liquid for collection in a three-necked flask, so D is correct.)

21.氮化钡 (Ba_3N_2) 是一种重要的化学试剂。高温下，向氢化钡 (BaH_2) 中通入氮气反应可制得氮化钡。已知：Ba_3N_2遇水反应；BaH_2在潮湿空气中能自燃，遇水反应。用如图所示装置制备氮化钡时，下列说法不正确的是 ( )(Barium nitride (Ba_3N_2) is an important chemical reagent. Barium nitride can be prepared by introducing nitrogen gas into barium hydride (BaH_2) at high temperature. Known: Ba_3N_2 reacts with water; BaH_2 can spontaneously ignite in humid air

and reacts with water. Which of the following statements is incorrect when preparing barium nitride with the device shown in the figure?)



A. 装置 B 中反应的化学方程式为 $NaNO_2+NH_4Cl\overset{\triangle}{=}N_2\uparrow+NaCl+2H_2O$（The chemical equation of the reaction in Device B is $NaNO_2 + NH_4Cl \overset{heating}{=} N_2\uparrow + NaCl + 2H_2O$.）

B. 气流由左向右的字母连接顺序为 c→a→b→d→e→g→f（The letter connection sequence of air flow from left to right is c→b→a→d→e→g→f.）

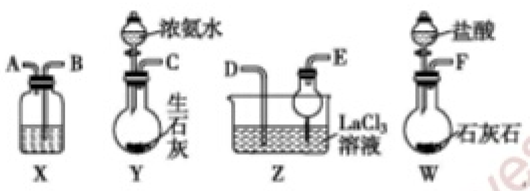
C. 实验时，先点燃装置 B 中的酒精灯，反应一段时间后，再点燃装置 C 中的酒精喷灯进行反应（During the experiment, first light the alcohol lamp in Device B, react for a period of time, and then light the alcohol blowtorch in Device C for reaction.）

D. 装置甲中的浓硫酸和装置丁中的碱石灰都是用于吸收水蒸气，防止水蒸气进入装置丙中（The concentrated sulfuric acid in Device A and soda lime in Device D are both used to absorb water vapor and prevent water vapor from entering Device C.）

答案（Answer）：B

解析（Analysis）：装置 B 用于制取 N_2，反应的化学方程式为 $NaNO_2+NH_4Cl\overset{\triangle}{=}N_2\uparrow+NaCl+2H_2O$，A 正确；装置 B 中产生的 N_2先干燥，再通入 C 中反应，故仪器连接顺序为 c→b→a→d→e→g→f，B 错误；实验时，先点燃装置 B 中的酒精灯，制取 N_2并排尽装置内的空气，再点燃装置 C 中的酒精喷灯进行反应，C 正确；由于 Ba_3N_2遇水反应，BaH_2在潮湿空气中能自燃，遇水反应，所以装置 A 和装置 D 的作用相同，都用于吸收水蒸气，防止水蒸气进入装置 C 中，D 正确。（Device B is used to prepare N_2, and the chemical equation of the reaction is $NaNO_2 + NH_4Cl \overset{heating}{=} N_2\uparrow + NaCl + 2H_2O$, so A is correct. The N_2 produced in Device B is dried first and then introduced into Device C for reaction, so the instrument connection sequence is c→b→a→d→e→g→f, so B is incorrect. During the experiment, first light the alcohol lamp in Device B to prepare N_2 and exhaust the air in the device, then light the alcohol blowtorch in Device C for reaction, so C is correct. Since Ba_3N_2 reacts with water and BaH_2 can spontaneously ignite in humid air and reacts with water, Device A and Device D have the same function, both absorbing water vapor and preventing water vapor from entering Device C, so D is correct.）

22.碳酸镧 [$La_2(CO_3)_3$] 常用于慢性肾衰患者高磷血症的治疗，实验室用如图所示装置，利用反应 $2LaCl_3+6NH_4HCO_3=La_2(CO_3)_3\downarrow+6NH_4Cl+3CO_2\uparrow+3H_2O$ 来制取。下列说法错误的是 ( )(Lanthanum carbonate [$La_2(CO_3)_3$] is often used in the treatment of hyperphosphatemia in patients with chronic renal failure. It is prepared in the laboratory with the device shown in the figure by the reaction $2LaCl_3 + 6NH_4HCO_3 = La_2(CO_3)_3\downarrow + 6NH_4Cl + 3CO_2\uparrow + 3H_2O$. Which of the following statements is incorrect?)



A. 装置 X 中盛放的试剂应为饱和 $NaHCO_3$溶液（The reagent contained in device X should be saturated $NaHCO_3$ solution.）

B. 实验中各装置导管口的连接方式为 F→B，A→D，E←C（The connection mode of each device nozzle in the experiment is F→B, A→D, E←C.)

C. 实验时可以不用 X 装置，将 W 中的盐酸用稀硫酸代替，实验效果相同（Device X cannot be omitted in the experiment, and dilute sulfuric acid cannot replace hydrochloric acid in W.)

D. 实验时，应先打开 Y 中分液漏斗的旋塞，一段时间后再打开 W 中分液漏斗的旋塞（During the experiment, first open the stopcock of the separatory funnel in Y, and then open the stopcock of the separatory funnel in W after a period of time.)

答案（Answer）：C

解析（Analysis）：盐酸易挥发，装置 X 的作用是除去二氧化碳中混有的 HCl 杂质，其中盛放的试剂应为饱和 $NaHCO_3$溶液，A 正确；装置 Y 用于制氨气，氨气极易溶于水，通入 $LaCl_3$溶液时要注意防倒吸，则 C 连 E，装置 W 用于制二氧化碳，制取的二氧化碳中混有 HCl 杂质，需通过装置 X 除去 HCl 气体，则 F 连接 B，A 连接 D，导管口的连接方式为 F→B，A→D，E←C，B 正确；装置 W 用于制取二氧化碳，将盐酸用稀硫酸代替，生成的 $CaSO_4$会附着在石灰石表面，不利于气体的产生，C 错误；实验开始时，先打开 Y 中分液漏斗的旋塞，先通入氨气，使 $LaCl_3$溶液呈碱性以吸收更多的二氧化碳，一段时间后再打开 W 中分液漏斗的旋塞，D 正确（Hydrochloric acid is volatile. The function of device X is to remove HCl impurities mixed in carbon dioxide, and the reagent contained in it should be saturated $NaHCO_3$ solution, so A is correct. Device Y is used to prepare ammonia gas, which is extremely soluble in water. Attention should be paid to anti-suck-back when introducing into $LaCl_3$ solution, so C is connected to E. Device W is used to prepare carbon dioxide, which is mixed with HCl impurities and needs to be removed by device X, so F is connected to B and A is connected to D. The nozzle connection mode is F→B, A→D, E←C, so B is correct. Device W is used to prepare carbon dioxide. If dilute sulfuric acid is used instead of hydrochloric acid, the generated $CaSO_4$ will adhere to the surface of limestone, which is not conducive to gas generation, so C is incorrect. At the beginning of the experiment, first open the stopcock of the separatory funnel in Y to introduce ammonia gas, make $LaCl_3$ solution alkaline to absorb more carbon dioxide, and then open the stopcock of the separatory funnel in W after a period of time, so D is correct.)

23.制取物质、探究物质的性质是学习化学必备的素养之一。实验室从含碘废液 (除水外含 CCl_4、I_2、I^-等) 中回收碘，其实验流程如下。下列说法正确的是 ( )(Preparing substances and exploring their properties is one of the essential qualities for learning chemistry. Iodine is recovered from iodine-containing waste liquid (containing CCl_4, I_2, I^-, etc. except water) in the laboratory, and the experimental process is as follows. Which of the following statements is correct?)

A. 操作 Ⅰ 为蒸馏（Operation Ⅰ is liquid separation.)

B. Na_2SO_3溶液的作用是还原剂（The function of Na_2SO_3 solution is a reducing agent.)

C. 物质 X 为硫酸钠（Substance X is CCl_4.)

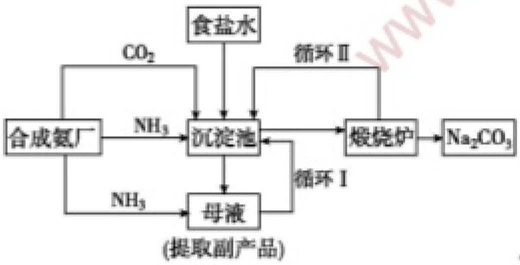
D. 上述流程中可能用到的仪器有分液漏斗、坩埚（The instruments possibly used in the above process include separatory funnel.)

答案（Answer）：B

解析（Analysis）：根据题意可知，加入 Na_2SO_3溶液，作为还原剂将碘还原成碘离子，离子方程式为 $I_2+SO_3^{2-}+H_2O=2I^-+SO_4^{2-}+2H^+$，含碘废液中除了水之外，还有 CCl_4，而 CCl_4与水溶液分层，则操作 Ⅰ 为分液，分液之后得到 CCl_4和含碘离子溶液，I^-被氧化剂 Y 氧化为 I_2。根据题意可知，含碘废液中除了水之外，还有 CCl_4，而 CCl_4与水溶液分层，则操作 Ⅰ 为分液，A 错误；根据题意可知，加入 Na_2SO_3溶液，作用是还原剂，将碘还原成碘离子，离子方程式为 $I_2+SO_3^{2-}+H_2O=2I^-+SO_4^{2-}+2H^+$，B 正

确；操作Ⅰ为分液，分液之后得到 CCl_4和含碘离子的溶液，物质 X 为 CCl_4，C 错误；流程中用不到坩埚，D 错误。（According to the question, adding Na_2SO_3 solution as a reducing agent reduces iodine to iodide ions, and the ionic equation is $I_2 + SO_3^{2-} + H_2O = 2I^- + SO_4^{2-} + 2H^+$. In addition to water, the iodine-containing waste liquid also contains CCl_4, which is layered with the aqueous solution, so operation Ⅰ is liquid separation. After liquid separation, CCl_4 and iodide ion-containing solution are obtained, and I^- is oxidized to I_2 by oxidant Y. According to the question, in addition to water, the iodine-containing waste liquid also contains CCl_4, which is layered with the aqueous solution, so operation Ⅰ is liquid separation, so A is incorrect. According to the question, adding Na_2SO_3 solution acts as a reducing agent to reduce iodine to iodide ions, and the ionic equation is $I_2 + SO_3^{2-} + H_2O = 2I^- + SO_4^{2-} + 2H^+$, so B is correct. Operation Ⅰ is liquid separation, and CCl_4 and iodide ion-containing solution are obtained after liquid separation. Substance X is CCl_4, so C is incorrect. Crucible is not used in the process, so D is incorrect.)

24.我国化学家侯德榜发明了联合制碱法，其主要过程如图所示。下列有关说法正确的是 ( ) (Chinese chemist Hou Debang invented the combined soda process, and its main process is shown in the figure. Which of the following statements is correct?)



A. CO_2先通入沉淀池，然后通入 NH_3（NH_3 is first introduced into the sedimentation tank, and then CO_2 is introduced.)

B. 沉淀池中发生反应的离子反应方程式为 $CO_2+NH_3+H_2O=HCO_3^-+NH_4^+$（The ionic equation of the reaction in the sedimentation tank is $Na^++CO_2+NH_3+H_2O=NaHCO_3\downarrow+NH_4^+$.)

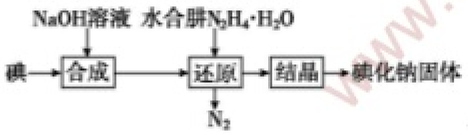
C. 循环Ⅰ可提高 Cl^-的利用率，循环Ⅱ中的物质是 CO_2（Cycle Ⅰ can improve the utilization rate of NaCl, and the substance in cycle Ⅱ is CO_2.)

D. 该流程的母液中提取的副产品为 NaCl（The by-product extracted from the mother liquor of this process is NH_4Cl.)

答案（Answer）：C

解析（Analysis）：联合制碱法是利用氯化钠、二氧化碳、氨气在水溶液中反应，生成碳酸氢钠，再煅烧碳酸氢钠获得碳酸钠的过程，由于二氧化碳的溶解度较小，因此往氯化钠溶液中通气体时应先通入溶解度较大的氨气，升高溶液的 pH，提高二氧化碳的溶解度；在沉淀池中沉淀出碳酸氢钠后，溶液中还有生成的氯化铵和过量的氯化钠，继续通氨气可提取出氯化铵副产品，同时母液中含有氯化钠，可循环利用；煅烧碳酸氢钠生成碳酸钠和二氧化碳，二氧化碳也可循环利用。因氨气在溶液中溶解度较大，二氧化碳溶解度较小，先通氨气使溶液呈碱性后再通二氧化碳，这样有利于二氧化碳吸收反应，试剂顺序不能颠倒，A 错误；沉淀池中 NaCl 溶液与 NH_3、CO_2反应生成碳酸氢钠和氯化铵，碳酸氢钠因为在水中溶解度较小而结晶析出，故反应的离子方程式为 $Na^++CO_2+NH_3+H_2O=NaHCO_3\downarrow+NH_4^+$，B 错误；循环Ⅰ可提高 NaCl 的利用率，循环Ⅱ中的 X 是 CO_2，C 正确；流程中母液中提取的副产品为 NH_4Cl，D 错误。（The combined soda process is a process of reacting sodium chloride, carbon dioxide and ammonia in aqueous solution to form sodium bicarbonate, and then calcining sodium bicarbonate to obtain sodium carbonate. Due to the low solubility of carbon dioxide, ammonia with high solubility should be introduced first when passing gas into sodium chloride solution to raise the pH of the solution and improve the solubility of carbon dioxide. After sodium bicarbonate precipitates in the sedimentation tank, the solution also contains generated ammonium chloride and excess sodium chloride. Continuing to pass ammonia can extract ammonium chloride by-product, and the mother liquor contains sodium chloride which can be recycled. Calcining sodium bicarbonate produces sodium carbonate and carbon dioxide, and carbon dioxide can also be recycled. Because ammonia has high solubility in solution and carbon dioxide has low solubility, ammonia is introduced first to make the solution alkaline and then carbon dioxide is introduced, which is conducive to the absorption reaction of carbon dioxide, and the reagent order cannot be reversed, so A is incorrect. NaCl solution reacts with NH_3 and CO_2 in the sedimentation tank to form sodium bicarbonate and ammonium chloride. Sodium bicarbonate crystallizes out due to its low solubility in water, so the ionic equation of the reaction is $Na^+ + CO_2 + NH_3 + H_2O = NaHCO_3\downarrow + NH_4^+$, so B is incorrect. Cycle Ⅰ can improve the utilization rate of NaCl, and X in cycle Ⅱ is CO_2, so C is correct. The by-product extracted from the mother liquor in the process is NH_4Cl, so D is incorrect.)

25.采用水合肼 ($N_2H_4 \cdot H_2O$) 还原法制取碘化钠固体，其制备流程如图所示。已知："合成" 步骤中生成的副产物为 IO_3^-。下列说法中不正确的是 ( )(Sodium iodide solid is prepared by hydrazine hydrate ($N_2H_4 \cdot H_2O$) reduction method, and its preparation process is shown in the figure. Known: the by-product generated in the synthesis step is IO_3^-. Which of the following statements is incorrect?)

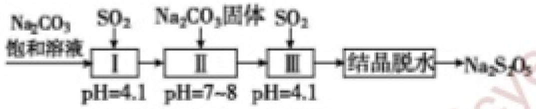


A. "合成" 过程所得溶液中主要含有 I^-、IO^-、IO_3^-和 Na^+（The solution obtained in the synthesis
B. "还原" 过程消耗 IO_3^-的离子方程式为 $2IO_3^-+3N_2H_4 \cdot H_2O=3N_2\uparrow+2I^-+9H_2O$（The ionic equation for consuming IO_3^- in the reduction process is $2IO_3^- + 3N_2H_4 \cdot H_2O = 3N_2\uparrow + 2I^- + 9H_2O$.)
C. "水合肼还原法" 的优点是 $N_2H_4 \cdot H_2O$ 被氧化后的产物为 N_2和 H_2O，不引入杂质（The advantage of hydrazine hydrate reduction method is that the oxidation products of $N_2H_4 \cdot H_2O$ are N_2 and H_2O without introducing impurities.)
D. 工业上常用铁屑还原 $NaIO_3$制备 NaI，理论上该反应中氧化剂与还原剂的物质的量之比为 1：2（The molar ratio of oxidant to reducing agent in this reaction is 1:3.)

答案（Answer）：D

解析（Analysis）："合成" 过程中 I_2与 NaOH 溶液主要发生反应：$I_2+2NaOH=NaI+NaIO+H_2O$，结合信息可知还有副产物 IO_3^-，则所得溶液中主要含有 I^-、IO^-、IO_3^-和 Na^+，A 正确；"还原" 过程水合肼将 IO_3^-还原为 I^-，水合肼则被氧化为 N_2，B 正确；由 B 项分析可知，$N_2H_4 \cdot H_2O$ 被氧化后的产物为 N_2和 H_2O，不引入杂质，C 正确；氧化剂：$NaIO_3 \rightarrow I^-\backslash 6e^-$，还原剂：$Fe \rightarrow Fe^{2+}\backslash 2e^-$，根据得失电子守恒可知，氧化剂与还原剂的物质的量之比为 1：3，D 错误。（In the synthesisprocess, I_2 mainly reacts with NaOH solution: $I_2 + 2NaOH = NaI + NaIO + H_2O$. Combined with the information, there is also a by-product IO_3^-, so the obtained solution mainly contains I^-, IO^-, IO_3^- and Na^+, so A is correct. In the reduction process, hydrazine hydrate reduces IO_3^- to I^-, and hydrazine hydrate is oxidized to N_2, so B is correct. According to the analysis of B, the oxidation products of $N_2H_4 \cdot H_2O$ are N_2 and H_2O without introducing impurities, so C is correct. Oxidant: $NaIO_3 \rightarrow I^-{\sim}6e^-$, reducing agent: $Fe \rightarrow Fe^{2+}{\sim}2e^-$. According to the conservation of electron gain and loss, the molar ratio of oxidant to reducing agent is 1:3, so D is incorrect.)

26.焦亚硫酸钠 ($Na_2S_2O_5$) 为白色或黄色结晶，有强烈的刺激性气味，溶于水，水溶液呈酸性，与强酸接触放出 SO_2并生成相应的盐类；久置空气中，易氧化变质。$Na_2S_2O_5$可由 $NaHSO_3$过饱和溶液经结晶脱水制得。利用 SO_2生产 $Na_2S_2O_5$的流程如图所示。下列相关叙述不正确的是 ( )(Sodium pyrosulfite ($Na_2S_2O_5$) is a white or yellow crystal with a strong pungent odor, soluble in water, and its aqueous solution is acidic. It releases SO_2 when in contact with strong acid and forms corresponding salts; it is easy to oxidize and deteriorate when placed in air for a long time. $Na_2S_2O_5$ can be prepared by crystallization and dehydration of $NaHSO_3$ supersaturated solution. The process of producing $Na_2S_2O_5$ from SO_2 is shown in the figure. Which of the following related statements is incorrect?)



A. 1 mol $Na_2S_2O_5$晶体中过氧键数目为 N_a（There is no peroxy bond in 1 mol $Na_2S_2O_5$ crystal.）

B. $Na_2S_2O_5$久置空气中可发生反应：$Na_2S_2O_5+O_2+H_2O=2NaHSO_4$（$Na_2S_2O_5$ can react in air for a long time: $Na_2S_2O_5 + O_2 + H_2O = 2NaHSO_4$.）

C. 反应 Ⅱ 为 $Na_2CO_3+2NaHSO_3=2Na_2SO_3+CO_2\uparrow+H_2O$（Reaction Ⅱ is $Na_2CO_3 + 2NaHSO_3 = 2Na_2SO_3 + CO_2\uparrow + H_2O$.）

D. 反应 Ⅲ 中充入 SO_2是为了得到 $NaHSO_3$过饱和溶液（Filling SO_2 in reaction Ⅲ is to obtain $NaHSO_3$ supersaturated solution.）

答案（Answer）：A

解析（Analysis）：$Na_2S_2O_5$晶体中 Na 元素为 + 1 价，S 元素为 + 4 价，O 元素为 - 2 价，故该晶体中无过氧键，故 A 错误；$Na_2S_2O_5$中 S 元素为 + 4 价，久置空气中可以被 O_2氧化为 + 6 价，发生反应：$Na_2S_2O_5+O_2+H_2O=2NaHSO_4$，故 B 正确；反应 Ⅱ 中加入 Na_2CO_3中和 $NaHSO_3$，化学方程式为 $Na_2CO_3+2NaHSO_3=2Na_2SO_3+CO_2\uparrow+H_2O$，故 C 正确；根据题意可知 $Na_2S_2O_5$可由 $NaHSO_3$过饱和溶液经结晶脱水制得，故反应 Ⅲ 中充入 SO_2是为了得到 $NaHSO_3$过饱和溶液，故 D 正确。（In $Na_2S_2O_5$ crystal, Na element is +1 valence, S element is +4 valence, and O element is -2 valence, so there is no peroxy bond in the crystal, so A is incorrect. S element in $Na_2S_2O_5$ is +4 valence, which can be oxidized to +6 valence by O_2 in air for a long time, and the reaction occurs: $Na_2S_2O_5 + O_2 + H_2O = 2NaHSO_4$, so B is correct. Na_2CO_3 is added to neutralize $NaHSO_3$ in reaction Ⅱ, and the chemical equation is $Na_2CO_3 + 2NaHSO_3 = 2Na_2SO_3 + CO_2\uparrow + H_2O$, so C is correct. According to the question, $Na_2S_2O_5$ can be prepared by crystallization and dehydration of $NaHSO_3$ supersaturated solution, so filling SO_2 in reaction Ⅲ is to obtain $NaHSO_3$ supersaturated solution, so D is correct.)